

Enhancing BioBERT with UMLS for Medical Question Answering

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Abstract

This study integrates the Unified Medical Language System (UMLS) with BioBERT to improve biomedical question answering (QA). UMLS components MRCONSO, MRSTY, and MRREL were used to augment the PubMedQA dataset with semantic types and relationships. Models were fine-tuned with and without UMLS integration, using dropout for regularization. Evaluation revealed that UMLS-enhanced models improved accuracy and F1-scores, with MRREL-augmented configurations demonstrating better generalization. The approach highlights the value of domain-specific knowledge in biomedical NLP and suggests future directions for multilingual and diverse datasets.

1 Introduction

Biomedical NLP tasks such as QA require understanding complex terminologies and relationships. While BioBERT excels in general biomedical text understanding, it lacks explicit domain-specific knowledge[4]. UMLS provides structured medical information, including concepts (MRCONSO), semantic types (MRSTY), and relationships (MRREL), which can enhance BioBERT’s performance. This study investigates the impact of integrating UMLS with BioBERT for QA on the PubMedQA dataset.

1.1 Problem Statement

Medical QA requires deep understanding of complex terminology and relationships. While BioBERT captures general patterns, it lacks domain-specific knowledge from ontologies like UMLS, limiting its effectiveness in answering intricate medical questions.

1.2 Motivation

- **Theoretical Merits:** UMLS enriches BioBERT with structured biomedical knowledge, enhancing semantic reasoning and interpretability.[5].
- **Practical Merits:** Improved QA performance aids clinical decisions, patient education, and research by delivering accurate answers.[3].

1.3 Research Question

How does the integration of UMLS components MRCONSO, MRSTY, and MRREL into BioBERT impact its performance on medical QA tasks?

2 Data and Preprocessing

2.1 Data

The PubMedQA dataset, consisting of 1,000 training, 200 validation, and 500 test samples labeled as *Yes*, *No*, or *Maybe*, was used for this study [2]. UMLS components were integrated to enhance the dataset, including MRCONSO (mapping terms to medical concepts), MRSTY (semantic types like *Disease* or *Procedure*), and MRREL (relationships such as *treats* or *causes*) [1] [5].

2.2 Preprocessing

The dataset was enriched with UMLS annotations to include concept, semantic type, and relationship information. Questions and contexts were tokenized using BioBERT’s tokenizer [4] [3], and final inputs combined tokenized text with UMLS-augmented embeddings for model training.

3 Methods

3.1 Baseline and UMLS-Augmented Models

The baseline BioBERT model was fine-tuned on the PubMedQA dataset without UMLS augmentation using cross-entropy loss and the Adam optimizer [3][devlin2019bert]. This served as a reference for evaluating UMLS-based enhancements.

To integrate UMLS, three models were developed: MRCONSO-augmented, incorporating concept-level embeddings; MRSTY-augmented, adding semantic type embeddings (e.g., *Disease*, *Procedure*); and MRREL-augmented, embedding relationships like *treats* or *causes* [5]. These models enriched BioBERT’s input representation with domain-specific knowledge.

Fine-tuning strategies included dropout regularization to mitigate overfitting and balance the enriched embeddings with generalization. Models were implemented using the Transformers library and trained with a batch size of 16 for 5-10 epochs. Performance was assessed using accuracy, precision, recall, and F1-score across training, validation, and test sets.

4 Results and Discussion

4.1 Quantitative Results

4.2 Discussion

Integrating UMLS into BioBERT significantly improved performance on training and validation sets[devlin2019bert], with the mrsty_carry model achieving the highest metrics. However, models like mrrel_carry and mrsty_carry showed overfitting, evidenced by performance drops on the test set.

The mrrel_fine_1 model demonstrated better generalization, balancing performance across datasets, making it a more reliable approach. To further improve generalization,

Model	Dataset	Accuracy	Precision	Recall	F1-Score
Baseline	Train	0.9075	0.9318	0.7274	0.7389
Baseline	Validation	0.6100	0.4038	0.4918	0.4395
Baseline	Test	0.6100	0.4215	0.4616	0.4399
mrsty	Train	0.8775	0.5962	0.6555	0.6239
mrsty	Validation	0.8775	0.5962	0.6555	0.6239
mrsty	Test	0.5900	0.3903	0.4283	0.4065
mrrel	Train	0.8775	0.9322	0.6584	0.6378
mrrel	Validation	0.6500	0.4242	0.5102	0.4632
mrrel	Test	0.6300	0.4182	0.4349	0.4199
mrsty_carry	Train	0.9963	0.9968	0.9882	0.9924
mrsty_carry	Validation	0.9963	0.9968	0.9882	0.9924
mrsty_carry	Test	0.5700	0.4407	0.4400	0.4397
mrrel_carry	Train	0.9975	0.9980	0.9922	0.9950
mrrel_carry	Validation	0.9975	0.9980	0.9922	0.9950
mrrel_carry	Test	0.5900	0.4806	0.4789	0.4757
mrrel_fine_1	Train	0.89250	0.942399	0.679823	0.669645
mrrel_fine_1	Validation	0.60000	0.391855	0.462010	0.423197
mrrel_fine_1	Test	0.64000	0.432304	0.432105	0.416031
model_mrsty_2	Train	0.99875	0.998765	0.996078	0.997409
model_mrsty_2	Validation	0.99875	0.998765	0.996078	0.997409
model_mrsty_2	Test	0.57000	0.450482	0.437080	0.440742
model_mrrel_2	Train	0.97875	0.979153	0.957743	0.967614
model_mrrel_2	Validation	0.60000	0.515731	0.494485	0.465000
model_mrrel_2	Test	0.59000	0.394841	0.422441	0.407876

Table 1: Performance Metrics for Baseline and UMLS-Augmented Models.

techniques like data augmentation, diverse datasets, and optimized training strategies are needed.

Despite these challenges, UMLS integration was scalable and feasible for large datasets, showing potential for broader applications in clinical decision support and biomedical research.

5 Conclusion, Challenges, and Limitations

5.1 Conclusion

Integrating UMLS with BioBERT improved medical question answering, showing significant gains on training and validation sets. These enhancements have promising applications in clinical decision support and biomedical research.

5.2 Challenges

The models faced limited generalization, with performance dropping on test data compared to training and validation. This highlights the need for more diverse datasets with a wider range of medical questions and contexts. Overfitting was also a challenge,

especially in UMLS-augmented models like MRSTY_{carry}, where enriched embeddings caused memorization of training data.

5.3 Limitations and Future Directions

Training UMLS-augmented models faced challenges such as overfitting, where models memorized training data but struggled with generalization to unseen data. GPU disk space issues arose from frequent saving of models and tokenizers, causing server crashes after training sessions. These limitations highlighted the need for more efficient storage and regularization strategies.

Future directions include expanding to multilingual datasets for global applicability and integrating additional ontologies like SNOMED CT to enrich understanding of medical concepts. Techniques such as increased dropout, early stopping, and data augmentation should be implemented to address overfitting. Transfer learning on smaller, domain-specific datasets can further improve scalability and performance for diverse medical question answering tasks.

6 References

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